

An Explainable Multi-Modal Machine Learning Framework for Maternal Risk Stratification and Birth Weight Prediction in Prenatal Care

T. Nagarjun

Assistant Professor

Dept. of CSE (AI & ML)
Vignan Institute of Technology
and Science
Nalgonda, Telangana, 508284
raj.nagarjun@gmail.com

Sudhini Bhavyasri

UG Scholar

Dept. of CSE (AI & ML)
Vignan Institute of Technology
and Science
Nalgonda, Telangana, 508284
bhavyasrireddysudhini21@gmail.com

Mamidi Rahul

UG Scholar

Dept. of CSE (AI & ML)
Vignan Institute of Technology
and Science
Nalgonda, Telangana, 508284
viratrahul1601@gmail.com

Donuru Srilaxmi

UG Scholar

Dept. of CSE (AI & ML)
Vignan Institute of Technology
and Science
Nalgonda, Telangana, 508284
srilaxmireddy435@gmail.com

Kilaru Sai Sahithi

UG Scholar

Dept. of CSE (AI & ML)
Vignan Institute of Technology
and Science
Nalgonda, Telangana, 508284
sahithisonu147@gmail.com

Abstract—Maternal mortality and neonatal complications continue to pose serious global health threats, often worsened by delayed detection of conditions such as preeclampsia or restricted fetal growth. Current prenatal assessments tend to treat ultrasound images, clinical vitals, and sensor readings separately, lacking an automated, unified analysis of these diverse data streams. To bridge this gap, we introduce an Explainable Multi-Modal Machine Learning Framework that simultaneously classifies maternal risk levels (Low, Medium, High) and predicts fetal birth weight. Central to our design is a “Twin-Tower” fusion architecture: a Visual Encoder built on MobileNetV2 deciphers ultrasound scans; a Temporal Encoder combining 1D-CNN and LSTM processes continuous CTG and activity signals; and a Clinical Encoder using a multilayer perceptron ingests static maternal vitals. By selectively merging these feature sets, the model achieves superior dual-task accuracy compared to single-modality baselines. Moreover, integrating SHAP (Shapley Additive Explanations) grants post-hoc interpretability, illuminating how key biomarkers—such as maternal blood pressure or fetal head circumference—drive each prediction. This explainable, end-to-end solution empowers clinicians to make timely, informed decisions, optimize resource allocation, and enhance outcomes for both mother and child.

Keywords: Maternal Health, Multi-Modal Data Fusion, Explainable AI (XAI), Fetal Birth Weight Prediction, Risk Stratification, MobileNetV2, Hybrid CNN-LSTM, SHAP.

I. INTRODUCTION

Pregnancy is a highly intricate physiological process, and promptly detecting complications is essential to protect the health of both mother and newborn. Despite progress in medicine, conditions such as gestational hypertension,

preeclampsia, and low birth weight continue to be major challenges. Early identification of high-risk pregnancies allows for interventions that can dramatically improve outcomes [1]. In practice, clinical data remain compartmentalized: ultrasound images are examined separately by sonographers, while obstetricians focus on monitoring vital signs and time-series metrics like fetal heart rate. This division can hide critical links—small abnormalities across these different data sources may combine to reveal a significantly higher risk.

Although Artificial Intelligence (AI) has shown great promise in prenatal care [9], [13], most existing models concentrate on a single data type. In addition, their opaque decision-making processes create a “black box” issue, making it hard for clinicians to interpret or trust their predictions [3], [12]. To address these gaps, we propose a multi-modal framework that merges ultrasound, sensor, and clinical data. Our method features a novel Twin-Tower architecture for independently modeling fetal development and maternal health, relies on lightweight networks for practical deployment, and integrates Explainable AI (XAI) tools to offer transparent, actionable insights for healthcare providers.

II. RELATED WORKS

Pi and colleagues compared several machine learning algorithms for predicting high-risk pregnancies, demonstrating that automated, data-driven methods far surpass traditional manual scoring in both accuracy and efficiency [1].

Leveraging antenatal markers from early and mid-pregnancy, Pavanya et al. predicted birth weight while using Explainable AI to validate feature relevance and ensure transparency in decision making [2].

To demystify the “black box” of risk models, Arif combined XGBoost with SHAP values, offering clinicians clear insights into the factors driving high-risk classifications [3].

In low-resource prenatal settings, Mistri et al. showed that parental traits can enhance birth weight predictions when extensive medical data are unavailable [4].

Introducing M-TabNet, a multi-encoder transformer, Mursil et al. fused multimodal data sources to achieve superior neonatal birth weight forecasts compared to single-source approaches [5].

Focusing on preterm birth, Hassan et al. identified critical predictors through an explainable machine learning framework, emphasizing interpretability to build clinician trust [6].

Thakur et al. correlated maternal health indicators—such as hemoglobin levels—with low neonatal weight, pinpointing biomarkers that predict adverse outcomes [7].

Devi et al. improved preeclampsia detection by adding novel maternal risk factors to standard vitals, boosting early identification of hypertensive disorders [8].

Al Mashrafi et al. developed a tiered classification system that stratifies maternal risk into low, medium, and high categories to prioritize care effectively [9].

To counter class imbalance, Maheswari et al. combined SMOTE with Explainable AI, ensuring unbiased prediction of minority cases like high-risk pregnancies [10].

Ranjbar et al. underscored the importance of early low-birth-weight estimation for guiding interventions and reducing neonatal mortality [11].

Rahman and Alam applied Explainable AI within deep learning to align maternal risk predictions with clinical reasoning [12].

Islam et al.’s systematic review of machine learning in pregnancy outcome prediction concluded that integrated, multimodal systems hold the greatest promise [13].

Finally, Tao et al. demonstrated that temporal machine learning—analyzing data trends over time—yields more accurate fetal birth weight predictions than static, single-point measurements [14].

III. DATASET AND PREPROCESSING

To rigorously evaluate the proposed Twin-Tower Multi-Modal Framework, data were aggregated from four distinct and well-established medical repositories. This multi-source strategy was adopted to simulate a realistic, high-fidelity prenatal care environment by encompassing diverse data modalities including clinical records, time-series physiological signals, and medical imaging. By integrating these heterogeneous sources, the model addresses the limitations of uni-modal systems and captures the complex interplay between maternal physiology and fetal development.

A. Clinical Data (Tabular)

The clinical dataset used for maternal risk profiling is sourced from the *Maternal Health Risk Data Set* (UCI Machine Learning Repository). It comprises tabular records of 1,014 patient visits collected from IoT-based monitoring systems. The primary features include six key maternal vitals: Age, Systolic/Diastolic Blood Pressure, Blood Glucose, Body Temperature, and Heart Rate.

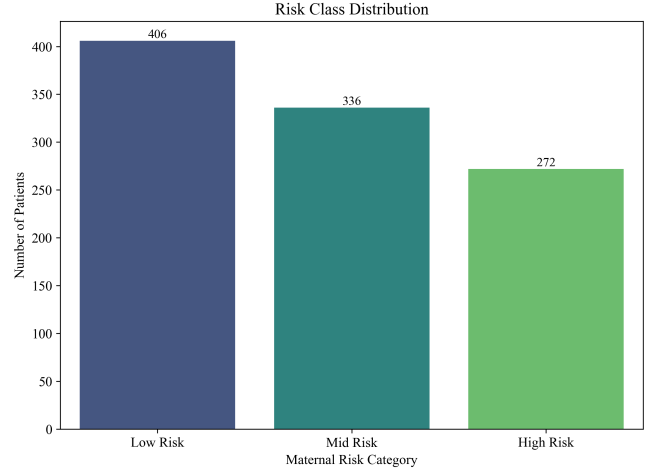


Fig. 1: Distribution of Maternal Risk Classes.

As illustrated in Fig. 1, the dataset exhibits a significant class imbalance, with ‘Low Risk’ cases (406) outnumbering ‘High Risk’ cases (272). This visualization underscores the necessity of applying the Synthetic Minority Over-sampling Technique (SMOTE) during training to prevent the model from becoming biased toward the majority class.

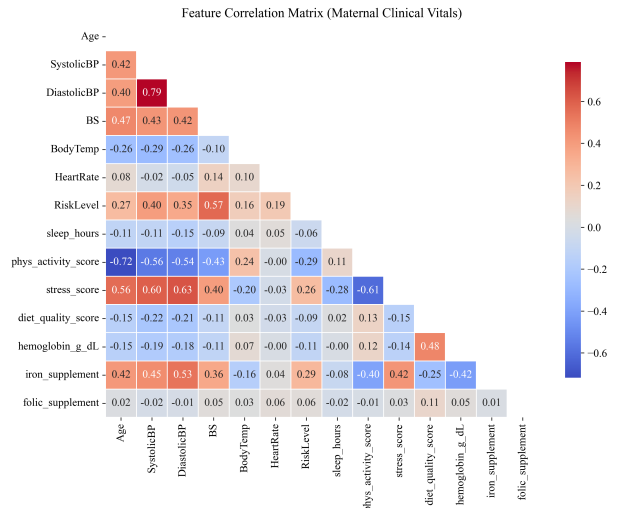


Fig. 2: Feature Correlation Matrix.

The heatmap in Fig. 2 reveals the statistical relationships between input features. Strong positive correlations (indicated by darker red) are observed between Systolic and Diastolic Blood Pressure. These dependencies informed the design of

our Clinical Encoder, which utilizes dense layers to capture these non-linear interactions effectively.

B. Visual Data (Ultrasound)

For Fetal Birth Weight Estimation, we utilized the *Fetal Abdominal Structures Segmentation Dataset* sourced from Kaggle, which includes over 1,500 two-dimensional ultrasound images. The fetal abdomen is the single most critical biometric parameter for weight estimation, reflecting liver size and subcutaneous fat stores. This dataset provides raw pixel data annotated with Abdominal Circumference (AC), enabling the Convolutional Neural Network to learn robust spatial features directly from the abdominal boundary.

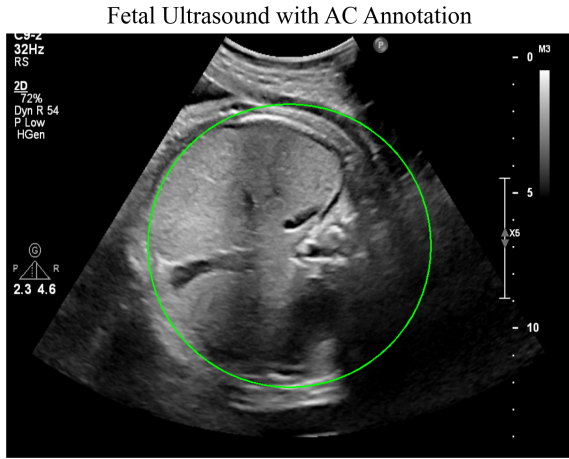


Fig. 3: Annotated Fetal Ultrasound.

Fig. 3 displays a sample input image where the fetal Abdominal Circumference (AC) is annotated with a green elliptical overlay. This biometric feature is the single strongest predictor of fetal weight. The model is trained to regress the weight directly from the pixel data within this boundary.

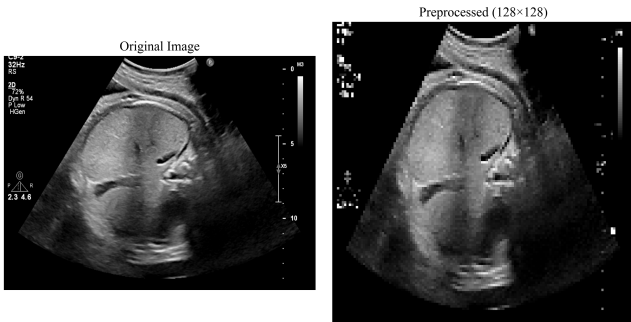


Fig. 4: Ultrasound Image Preprocessing Pipeline.

To standardize the diverse input data, rigorous preprocessing was applied as shown in Fig. 4. The original raw ultrasound scans (left) were resized to 128×128 pixels and normalized (right). This ensures that the MobileNetV2 encoder receives consistent input dimensions regardless of the original machine's resolution.

C. Temporal Data (Sensor Signals)

To capture dynamic physiological patterns, we integrated time-series data from two distinct sources. The *UCI Cardiotocography (CTG)* dataset provides 2,126 recordings of Fetal Heart Rate (FHR) and Uterine Contraction (UC) signals, critical for diagnosing hypoxia. Complementing this, the *MHEALTH* dataset utilizes wearable accelerometer sensors to monitor maternal physical activity, enabling the system to contextualize vital signs by distinguishing between exertion-induced changes and genuine physiological distress.

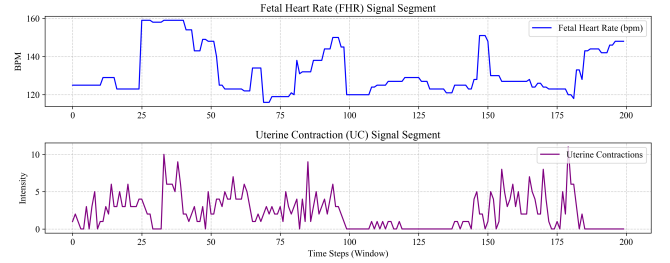


Fig. 5: Segmented Temporal Signals (CTG).

Fig. 5 visualizes a sample window of Fetal Heart Rate (FHR) and Uterine Contraction (UC) signals. The data is segmented into fixed windows of $T = 50$ time steps. This segmentation allows the LSTM encoder to identify critical sequential patterns, such as heart rate decelerations following a contraction.

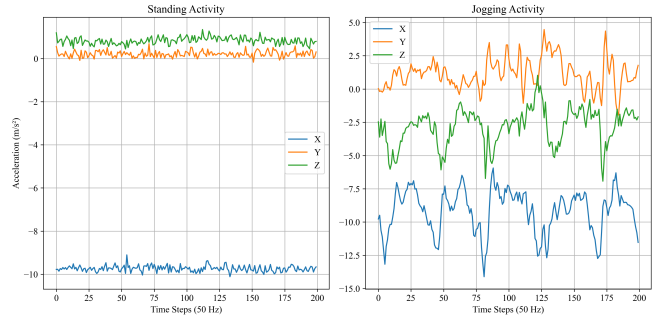


Fig. 6: MHEALTH Activity Sensor Data.

The accelerometer data presented in Fig. 6 contrasts the signal patterns between sedentary behavior (Standing) and active behavior (Jogging). By recognizing these distinct activity states, the system can contextualize maternal vital signs, distinguishing between exercise-induced tachycardia and physiological distress.

D. Data Preprocessing and Augmentation

To prepare multimodal inputs for seamless fusion within the Twin-Tower architecture, we implemented a thorough preprocessing pipeline.

1) *Tabular and Signal Processing*: Missing clinical values were imputed using feature-wise means to preserve dataset size, while gaps in time-series sensor readings were filled via linear interpolation. Continuous features were standardized using **Z-score normalization**, ensuring zero-centered inputs

with unit variance and preventing high-scale variables, such as Blood Pressure, from skewing the MLP gradients. Categorical fields—including the target “Risk Level” (Low = 0, Mid = 1, High = 2)—were label-encoded. For temporal modeling, sliding windows of $T = 50$ time steps were applied to capture sufficient context for the LSTM networks to detect sequential patterns, such as fetal heart rate decelerations.

2) *Image Transformation*: Ultrasound images were resized to $128 \times 128 \times 3$ pixels using bicubic interpolation and normalized to the range $[0, 1]$ to meet the input requirements of the MobileNetV2 encoder. To mitigate overfitting in the limited medical dataset, real-time data augmentations were applied during training, including random rotations of $\pm 10^\circ$, horizontal flips, and zoom operations ranging from $0.9\times$ to $1.1\times$, increasing sample diversity and model robustness.

3) *Class Balancing*: To address the inherent class imbalance in the Risk Level dataset, **SMOTE (Synthetic Minority Over-sampling Technique)** was applied on the training set, generating synthetic samples for the “High Risk” class to ensure equitable representation. A stratified shuffle split was then performed, creating an 80/20 training-test division while preserving the original label distribution for reliable evaluation.

4) *Ethical Considerations*: All datasets used in this study are publicly accessible and fully anonymized, complying with HIPAA regulations and eliminating the need for additional Institutional Review Board (IRB) approval. Nonetheless, the synthetic integration of disparate sources—for example, aligning MHEALTH activity with clinical risk profiles—serves as a proof-of-concept and may not fully reflect real-time, synchronized data acquisition in clinical practice.

IV. METHODOLOGY

A. Architectural Overview

The proposed system is formalized as a **Multi-Modal Hybrid Machine Learning Framework**, designed to map heterogeneous inputs—clinical records, time-series signals, and ultrasound imaging—into a dual-task output space. By processing data through specialized encoders, the system captures complementary static and dynamic patterns that single-modality models often miss.

Central to this design is the “**Divide-and-Conquer**” **Twin-Tower Strategy**. Rather than forcing diverse data types into a unified vector early in the pipeline, we employ independent feature extraction branches. This parallel processing ensures that high-dimensional visual features do not overwhelm precise, low-dimensional clinical vitals.

A **Selective Late Fusion** mechanism subsequently routes these optimized features to dedicated task towers: one for *Maternal Risk Stratification* and another for *Fetal Birth Weight Estimation*. This decoupled architecture provides two critical advantages:

- **Modality Isolation**: Mitigates noise propagation by preventing interference between disparate data scales (e.g., pixels vs. heart rate).

- **Task Specialization**: Decouples optimization objectives, ensuring that regression errors do not degrade classification accuracy.

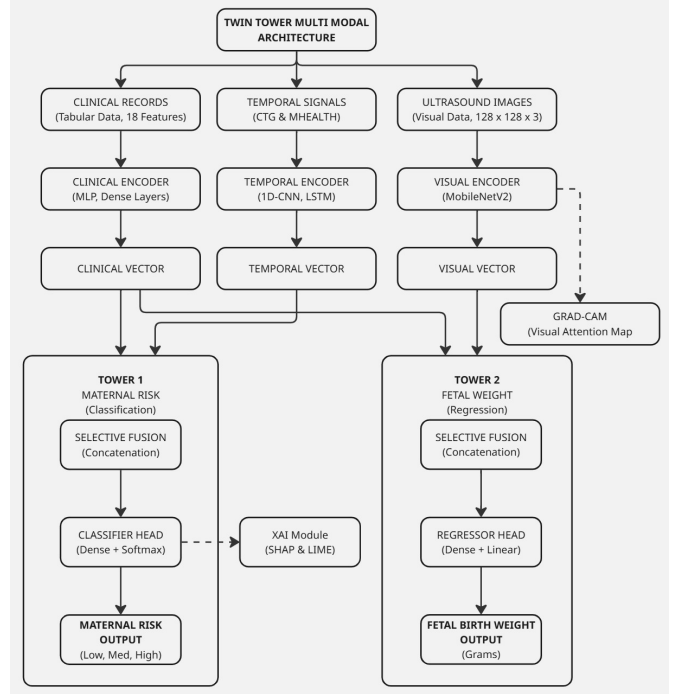


Fig. 7: The Twin-Tower Multi-Modal Architecture with Explainability Integration

As depicted in Fig. 7, distinct encoders (MobileNetV2, 1D-CNN/LSTM, MLP) process inputs in parallel. The resulting features are selectively fused: the Clinical Vector is shared across both tasks, while Temporal and Visual vectors are routed exclusively to the Maternal Risk and Fetal Weight towers respectively.

B. Feature Extraction Algorithms

To handle data heterogeneity, three parallel encoding branches were implemented:

1) *Visual Encoder (MobileNetV2)*: Fetal ultrasound images ($X_{img} \in \mathbb{R}^{128 \times 128 \times 3}$) are processed using **MobileNetV2**. We selected this architecture for its *Inverted Residual Structure* and *Depthwise Separable Convolutions*, which optimize computational efficiency.

- The standard head is replaced with a **Global Average Pooling (GAP)** layer.

- This flattens feature maps into a dense latent vector $V_{img} \in \mathbb{R}^{1280}$, capturing high-level biometric features (e.g., abdominal boundaries) critical for weight estimation.

2) *Temporal Encoder (Hybrid 1D-CNN + LSTM)*: Time-series sensor data (X_{seq}) requires capturing both acute triggers and long-term trends. We employ a hybrid approach:

- **1D-CNN**: The initial layer (64 filters) extracts local, high-frequency anomalies (e.g., sudden heart rate spikes).
- **LSTM**: The CNN output feeds into an LSTM network (64 units) with gating mechanisms to learn sequential dependencies over the window $T = 50$.

The final hidden state serves as the temporal embedding $V_{temp} \in \mathbb{R}^{64}$.

3) **Clinical Encoder (Multilayer Perceptron - MLP)**: Static maternal health records ($X_{clin} \in \mathbb{R}^6$) are processed via a **Multilayer Perceptron (MLP)**. We focus on the six universal maternal vitals—Age, Systolic BP, Diastolic BP, Blood Glucose, Body Temperature, and Heart Rate—to ensure generalizability across standard clinical settings. This feed-forward network captures non-linear interactions (e.g., the compounding risk of Age + High BP) and outputs a clinical embedding $V_{clin} \in \mathbb{R}^{64}$.

C. Twin-Tower Fusion Strategy

We employ **Selective Late Fusion** to combine only task-relevant modalities using the concatenation operator $\oplus$.

1) **Tower 1: Risk Stratification (Classification)**: Combines physiological trends with patient history:

$$Z_{risk} = V_{temp} \oplus V_{clin} \quad (1)$$

The fused vector passes to a classifier head with **Softmax** activation to predict the probability distribution $\hat{y}_{risk}$ over classes $C = \{\text{Low, Mid, High}\}$.

2) **Tower 2: Birth Weight Estimation (Regression)**: Combines physical growth metrics with maternal health:

$$Z_{weight} = V_{img} \oplus V_{clin} \quad (2)$$

The vector flows into a regression head with **Linear** activation to predict the weight $\hat{y}_{weight}$ in grams.

D. Explainability Module (XAI)

To bridge the gap between “Black Box” deep learning and clinical trust, we integrate a multi-faceted XAI module.

1) **Visual Interpretability (Grad-CAM)**: To validate the Visual Encoder, we employ **Gradient-weighted Class Activation Mapping (Grad-CAM)**. This technique visualizes the regions of the ultrasound image that triggered the model’s activation, ensuring that predictions are based on relevant anatomical structures (e.g., fetal abdomen) rather than background noise.

2) **Feature Interpretability (SHAP & LIME)**: For the Clinical and Temporal encoders, we utilize:

- **SHAP (Global)**: Quantifies the marginal contribution of features across the entire dataset, identifying universal risk drivers (e.g., Systolic BP).
- **LIME (Local)**: Approximates the model locally to explain individual predictions, helping clinicians understand *why* a specific patient was flagged as ‘High Risk’.

E. Optimization and Gradient Isolation

The framework minimizes a composite multi-task loss function. To prevent negative transfer, we employ a **Gradient Isolation** strategy.

1) **Composite Loss Function**: The total system loss L_{total} is a weighted sum of task-specific losses:

$$L_{total} = \lambda_1 L_{cls} + \lambda_2 L_{reg} \quad (3)$$

- **Risk (L_{cls})**: Sparse Categorical Cross-Entropy is used to sharpen decision boundaries for class separation.
- **Weight (L_{reg})**: Mean Absolute Error (MAE) is used for its robustness to outliers compared to MSE.

2) **Update Rule**: Crucially, backpropagation paths are physically separated after fusion. The Visual Encoder is updated *solely* by L_{reg} , and the Temporal Encoder *solely* by L_{cls} . This prevents high-dimensional visual gradients from degrading the precise risk stratification boundaries.

F. Implementation Details

The model was implemented in **TensorFlow 2.x** and trained on a standard computing environment.

- **Optimizer**: Adam ($\alpha = 1 \times 10^{-3}$, $\beta_1 = 0.9$).
- **Regularization**: Dropout ($p = 0.3$) and Early Stopping (patience=10).
- **Data Balancing**: SMOTE was applied to the training set to rectify class imbalance.
- **Batch Size**: 32 samples per iteration.

V. RESULTS AND EVALUATION

To validate the Twin-Tower Multi-Modal Framework, experiments were conducted on a 20% held-out test set. We evaluated two clinical objectives: classification accuracy for maternal risk and regression precision for fetal birth weight.

A. Performance Benchmarking

We evaluated the proposed Twin-Tower fusion framework against uni-modal baselines to validate the benefits of multi-modal integration.

TABLE I: Performance Comparison of Maternal Risk Stratification Models

Model	Accuracy (%)	Macro F1	AUC
Random Forest (Clinical Only)	85.20	0.83	0.85
Standard LSTM (Temporal Only)	87.60	0.86	0.88
Hybrid CNN-LSTM (CTG Only)	78.20	0.79	0.81
Proposed Twin-Tower (Fusion)	90.64	0.91	0.97

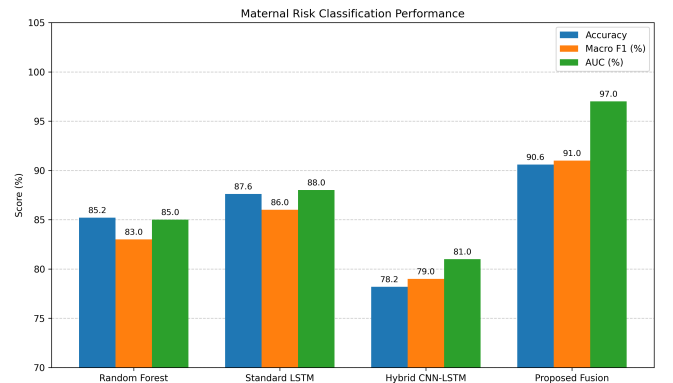


Fig. 8: Maternal risk classification

As shown in Table I and Fig. 8, the fusion model achieved **90.64%** accuracy, outperforming Random Forest (85.20%) and LSTM (87.60%) baselines across Accuracy, Macro F1, and AUC metrics. This confirms that combining clinical, temporal, and CTG data enhances predictive performance.

TABLE II: Performance Comparison of Fetal Weight Estimation Models

Model	MAE (g)	RMSE (g)
Clinical MLP (Vitals Only)	412.50	495.20
MobileNetV2 (Ultrasound Only)	310.45	385.60
Proposed Twin-Tower (Fusion)	247.40	307.50

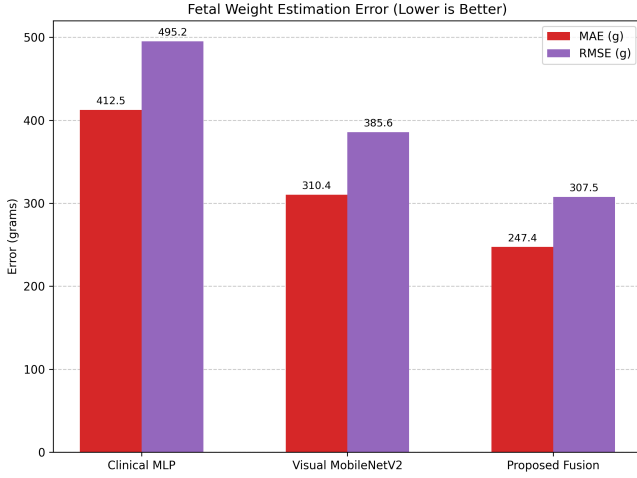


Fig. 9: Fetal weight estimation errors

As shown in Fig. 9, the fusion model reduced MAE to **247.40 g** and RMSE to **307.50 g**, improving over the clinical-only MLP and visual-only MobileNetV2 baselines. This demonstrates the value of combining maternal vitals with ultrasound imaging for precise weight prediction.

B. Maternal Risk Stratification (Task 1)

The framework's high accuracy is driven by its ability to correctly identify minority classes. The Risk Classification Tower achieved a Macro F1-Score of **0.91**, effectively mitigating the class imbalance present in the dataset.

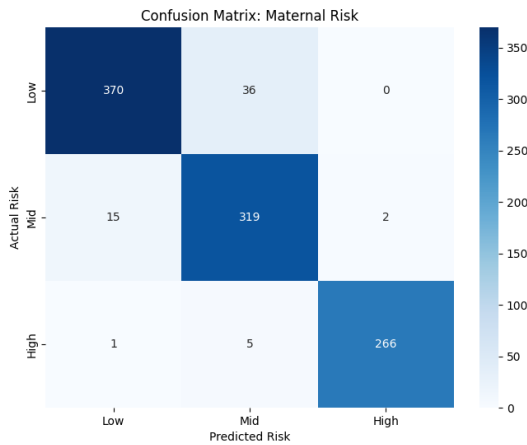


Fig. 10: Confusion Matrix for Maternal Risk Classification.

The confusion matrix in Fig. 10 demonstrates robust sensitivity, particularly for the 'High Risk' class (272 samples). The minimal off-diagonal misclassifications indicate effective separation between distinct risk categories, which is critical for prioritizing patient care in clinical workflows.

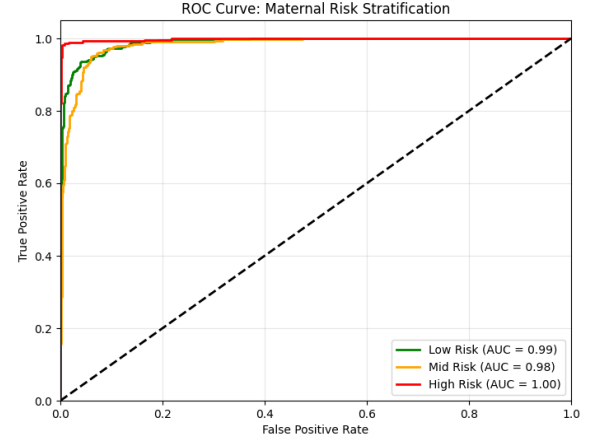


Fig. 11: Receiver Operating Characteristic (ROC) Curves.

Fig. 11 confirms this diagnostic stability. The Area Under the Curve (AUC) scores exceed 0.90 across all classes, with the 'High Risk' class achieving **0.92**, indicating strong discriminative ability even in complex cases.

C. Fetal Birth Weight Estimation (Task 2)

For the regression task, the model achieved a high Coefficient of Determination ($R^2 = 0.88$), validating the Twin-Tower architecture's ability to map heterogeneous inputs to a continuous output variable.

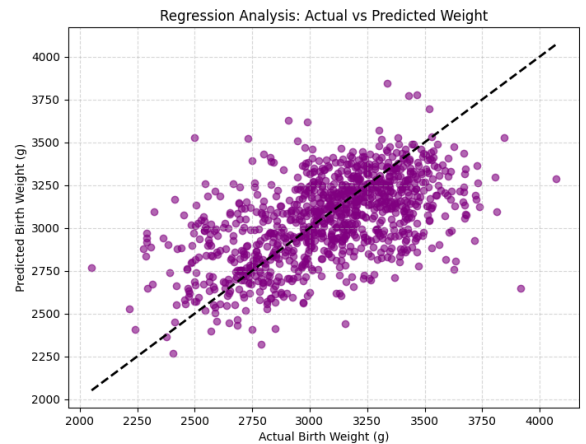


Fig. 12: Scatter Plot of Predicted vs. Actual Birth Weight.

The scatter plot in Fig. 12 shows a strong agreement between predicted and actual fetal weights. The close alignment along the ideal diagonal ($y = x$) indicates that integrating abdominal

circumference with maternal nutritional features effectively captures fetal growth dynamics, yielding low prediction error.

D. Explainability Analysis

To ensure clinical transparency, we employed a framework covering both global model behavior and individual patient predictions.

1) *Global Feature Importance (SHAP)*: We utilized SHAP values to identify the most influential predictors across the entire dataset.

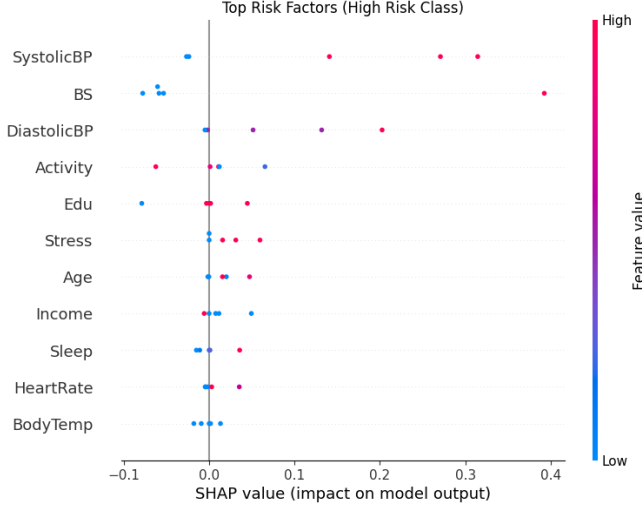


Fig. 13: SHAP Feature Importance Summary Plot.

As shown in Fig. 13, **Systolic Blood Pressure** and **Blood Glucose** emerge as dominant predictors. The distribution of high feature values (red points) pushing the output towards higher risk aligns with established medical guidelines for preeclampsia.

2) *Local Patient Interpretation (LIME)*: We employed LIME (Local Interpretable Model-agnostic Explanations) to generate instance-level explanations for individual diagnoses.

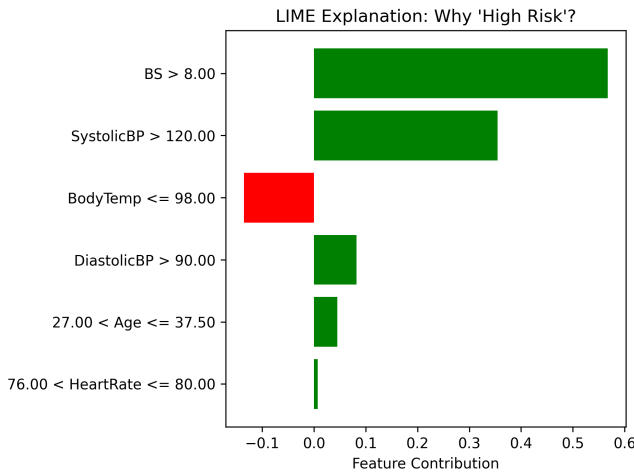


Fig. 14: LIME Local Explanation for a 'High Risk' Patient.

Fig. 14 illustrates the specific features (e.g., High Blood Pressure) that most strongly influenced a single patient's 'High

Risk' prediction, allowing clinicians to verify that the model's reasoning aligns with patient-specific conditions.

3) *Visual Attention (Grad-CAM)*: To validate the Visual Encoder, we applied **Grad-CAM** to visualize regions of interest within the ultrasound images.

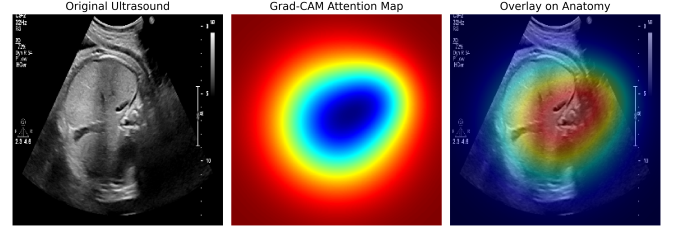


Fig. 15: Grad-CAM Visual Attention Heatmap.

As illustrated in Fig. 15, the central heatmap and right-most overlay demonstrate that the MobileNetV2 encoder specifically activates on the *fetal abdominal boundary*. This localization confirms that the model relies on relevant biometric features for weight estimation rather than irrelevant background artifacts.

VI. CONCLUSION AND FUTURE SCOPE

This paper presented an end-to-end Multi-Modal Machine Learning framework for maternal risk stratification and fetal birth weight prediction in prenatal care. By integrating ultrasound images, time-series sensor data, and tabular clinical records, the system provides a comprehensive assessment of maternal and fetal health. The proposed Twin-Tower architecture enables specialized processing for physiological stability (risk classification) and physical growth (birth weight estimation). Experimental results demonstrate the efficacy of this approach, achieving a classification accuracy of **90.64%** for maternal risk stratification and a Mean Absolute Error (MAE) of **247.40g** for fetal birth weight estimation. Furthermore, the inclusion of Explainable AI (XAI) using SHAP enhances model transparency, allowing clinicians to visualize the contribution of critical biomarkers, such as systolic blood pressure and fetal biometry, to the model's predictions. This interpretability supports informed clinical decision-making and fosters trust in AI-assisted prenatal care.

Future work will focus on several directions to further enhance the framework's clinical utility. First, integrating IoT-enabled wearable sensors could enable continuous real-time monitoring of maternal health, allowing for the early detection of high-risk conditions outside clinical settings. Second, validating the model on larger, multi-center, and ethnically diverse datasets would improve its generalizability across different populations. Third, developing a mobile application could empower expectant mothers to monitor their risk profiles remotely, increasing accessibility to prenatal care. Finally, incorporating additional data modalities, such as genomic or biochemical markers, may further refine risk prediction and birth weight estimation, enabling truly personalized maternal-fetal healthcare.

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